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GATE ECE · DIGITAL CIRCUITS · CHAPTER-WISE PYQS

Combinational Circuits

GATE ECE Digital Circuits · Previous Year Questions with Answer Key

45	36	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 2	23 questions
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Marks 8	1 question
Marks 10	4 questions
Answer key	at the end

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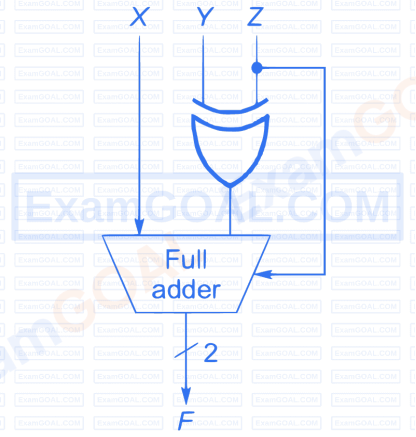
Marks 1 16 questions

Q1. GATE ECE 2025

Open solution

A full adder and an XOR gate are used to design a digital circuit with inputs X, Y, and Z, and output F, as shown below. The input Z is connected to the carry-in input of the full adder.

If the input Z is set to logic ' 1 ', then the circuit functions as _____ with X and Y as inputs.



(A) an adder

(B) a subtractor

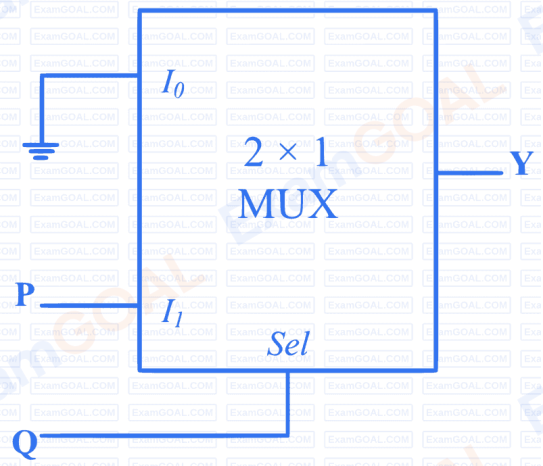
(C) a multiplier

(D) a binary to Gray code converter

Q2. GATE ECE 2023

Open solution

In the circuit shown below, P and Q are the inputs. The logical function realized by the circuit shown below is



(A) Y = PQ

(B) Y = P + Q

(C) Y = $\overline{PQ}$

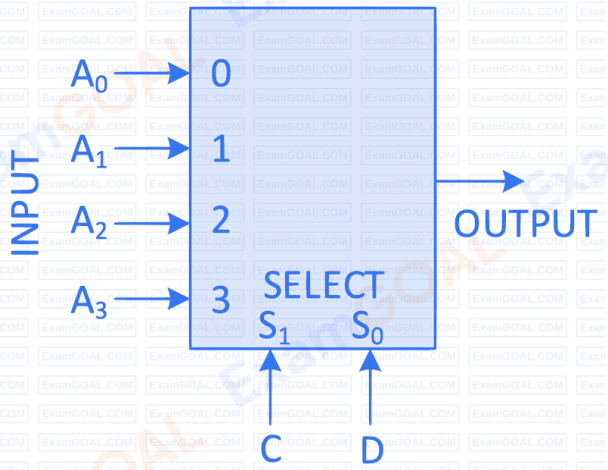
(D) Y = $\overline{P + Q}$

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Q3. GATE ECE 2022

Open solution

Consider the 2-bit multiplexer (MUX) shown in the figure. For OUTPUT to be the XOR of C and D, the values for A_0, A_1, A_2 and A_3 are _____.



(A) $A_0 = 0, A_1 = 0, A_2 = 1, A_3 = 1$

(B) $A_0 = 1, A_1 = 0, A_2 = 1, A_3 = 0$

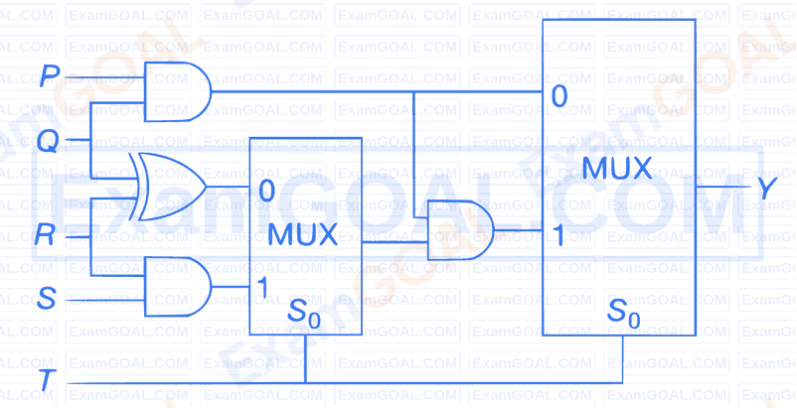
(C) $A_0 = 0, A_1 = 1, A_2 = 1, A_3 = 0$

(D) $A_0 = 1, A_1 = 1, A_2 = 0, A_3 = 0$

Q4. GATE ECE 2021

Open solution

The propagation delays of the XOR gate, AND gate and multiplexer (MUX) in the circuit shown in the figure are $4\ \text{ns}, 2\ \text{ns}$ and 1 ns respectively.



If all the inputs P, Q, R, S and T are applied simultaneously and held constant, the maximum propagation delay of the circuit is

(A) 3 ns

(B) 6 ns

(C) 5 ns

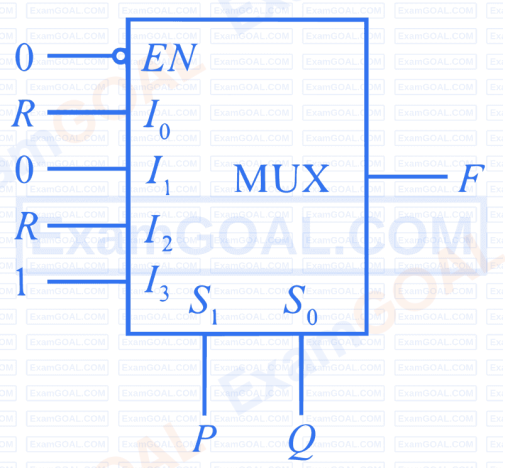
(D) 7 ns

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Q5. GATE ECE 2020

Open solution

The figure below shows a multiplexer where S_1 and S_0 are the select lines, I_0 to I_3 are the input data lines, EN is the enable line, and $F(P, Q, R)$ is the output. F is

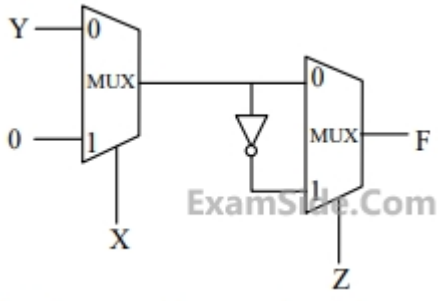


(A) $\bar{Q} + PR$ (B) $P\bar{Q}R + \bar{P}Q$ (C) $P + Q\bar{R}$ (D) $PQ + \bar{Q}R$

Q6. GATE ECE 2017 Set 2

Open solution

Consider the circuit shown in the figure.



The Boolean expression F implemented by the circuit is

(A) $\overline{X}\,\overline{Y}\,\overline{Z} + XY + \overline{Y}\,Z$ (B) $\overline{X}\,Y\,\overline{Z} + XZ + \overline{Y}\,Z$

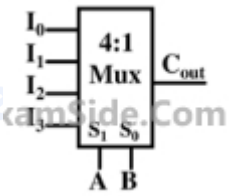
(C) $\overline{X}\,Y\,\overline{Z} + XY + \overline{Y}\,Z$ (D) $\overline{X}\,\overline{Y}\,\overline{Z} + XZ + \overline{Y}\,Z$

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Q7. GATE ECE 2016 Set 2

Open solution

A 4:1 multiplexer is to be used for generating the output carry of a full adder. A and B are the bits to be added while C_{in} is the input carry and C_{out} is the output carry. A and B are to be used as the select bits with A being the more significant select bit.



Which one of the following statements correctly describes the choice of signals to be connected to the inputs I_0, I_1, I_2 and I_3 so that the output is C_{out}?

(A) $I_0 = 0, I_1 = C_{in}, I_2 = C_{in}\, and\, I_3 = 1$

(B) $I_0 = 1, I_1 = C_{in}, I_2 = C_{in}\, and\, I_3 = 1$

(C) $I_0 = C_{in}, I_1 = 0, I_2 = 1\, and\, I_3 = C_{in}$

(D) $I_0 = 0, I_1 = C_{in}, I_2 = 1\, and\, I_3 = C_{in}$

Q8. GATE ECE 2014 Set 2

Open solution

In a half-subtractor circuit with X and Y as inputs, the Borrow (M) and Difference (N = X - Y) are given by

(A) $M = X \oplus Y, N = XY$

(B) $M = XY,\ N = X \oplus Y$

(C) $M = \overline{X}Y,\ N = X \oplus Y$

(D) $M = X\overline{Y},\ N = \overline{X \oplus Y}$

Q9. GATE ECE 2012

Open solution

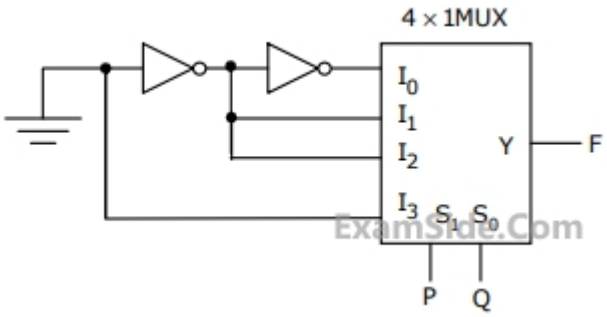
The output Y of a 2-bit comparator is logic 1 whenever the 2-bit input A is greater than the 2-bit input B. The number of combinations for which the output is logic 1, is

(A) 4 (B) 6 (C) 8 (D) 10

Q10. GATE ECE 2011

Open solution

The logic function implemented by the circuit below is (ground implies a logic "0"



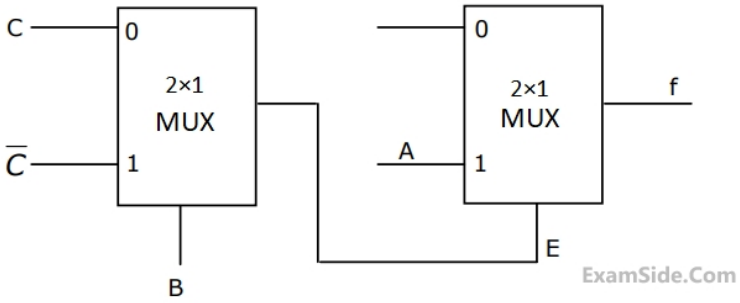
(A) F= AND (P, Q) (B) F= OR (P, Q) (C) F= XNOR (P, Q) (D) F= XOR (P, Q)

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Q11. GATE ECE 2005

Open solution

The Boolean function f implemented in the figure using two input multiplexers is



(A) $A\overline{B}C + AB\overline{C}$
(B) $ABC + A\overline{B}\overline{C}$
(C) $\overline{A}BC + \overline{A}\overline{B}\overline{C}$
(D) $\overline{A}\overline{B}C + \overline{A}B\overline{C}$

Q12. GATE ECE 2003

Open solution

Without any additional circuitry, an 8:1 MUX can be used to obtain

(A) some but not all Boolean functions of 3 variables
(B) all functions of 3 variables but none of 4 variables
(C) all functions of 3 variables and some but not all of 4 variables
(D) all functions of 4 variables and some but not all of 4 variables

Q13. GATE ECE 1997

Open solution

A 2-bit binary multiplier can be implemented using

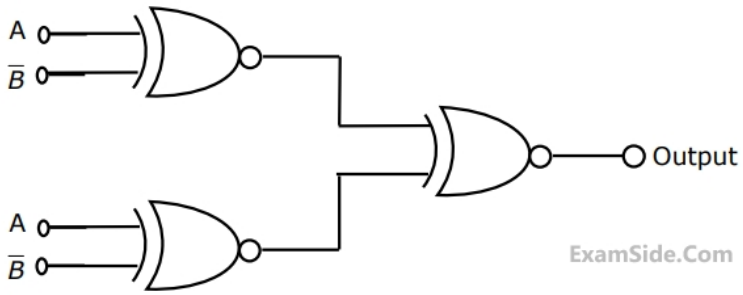
(A) 2 input AND gates only.
(B) 2 number of 2-input XOR gates and 6 number of 2-input AND gates.
(C) Two 2-input NOR gates and one XNOR gate.
(D) XOR gates and shift registers.

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Q14. GATE ECE 1995

The output of the circuit shown in figure is equal to

Open solution



(A) 0

(B) 1

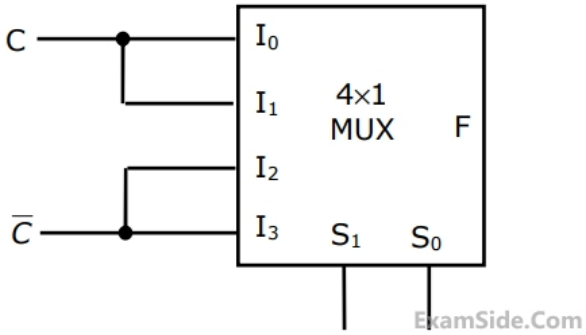
(C) $\overline{A}B + A\overline{B}$

(D) $\left(\overline{A \oplus B}\right) \oplus \left(\overline{A \oplus B}\right)$

Q15. GATE ECE 1992

The logic realized by the circuit shown in figure is

Open solution



(A) F = A ⊙ C (B) F = A ⊕ C (C) F = B ⊙ C (D) F = B ⊕ C

Q16. GATE ECE 1990

The minimum function that can detect a "divisible by 3" 8421 BCD code digit (representation is D_8 D_4 D_2 D_1) is given by:

Open solution

(A) $D_8 D_1 + D_4 D_2 + \overline{D_8} D_2 D_1$

(B) $D_8 D_1 + D_4 D_2 \overline{D_1} + \overline{D_4} D_2 D_1 + \overline{D_8}\,\overline{D_4}\,\overline{D_2}\,\overline{D_1}$

(C) $D_8 D_1 + D_4 D_2 + \overline{D_8}\,\overline{D_4}\,\overline{D_2}\,\overline{D_1}$

(D) $D_4 D_2 \overline{D_1} + D_4 D_2 D_1 + D_8 \overline{D_4} D_2 D_1$

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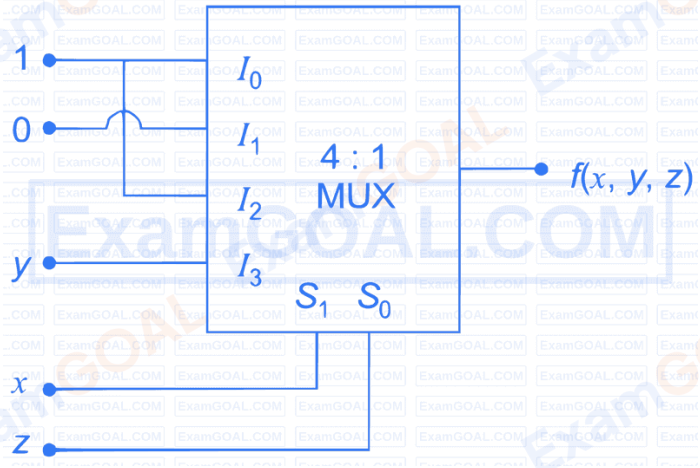
Marks 2 23 questions

Q17. GATE ECE 2026

A Boolean function, $f(x, y, z)$ with x as MSB and z as LSB is realized by $4:1$ multiplexer (MUX) with select lines, S_1 and S_0 (S_1 is MSB, S_0 is LSB) and inputs, I_0, I_1, I_2, I_3 as shown in the Figure.

Which of the following options is the correct expression of $f(x, y, z)$?

Open solution



(A) $xz + y$ (B) $x\bar{y} + z$ (C) $xy + \bar{z}$ (D) $\bar{x}y + \bar{z}$

Q18. GATE ECE 2024

A 4-bit priority encoder has inputs D_3, D_2, D_1, and D_0 in descending order of priority. The two-bit output AB is generated as 00, 01, 10, and 11 corresponding to inputs D_3, D_2, D_1, and D_0, respectively. The Boolean expression of the output bit B is _____.

Open solution

(A) $\overline{D_3}\,\overline{D_2}$ (B) $\overline{D_3}D_2 + \overline{D_3}\,\overline{D_1}$

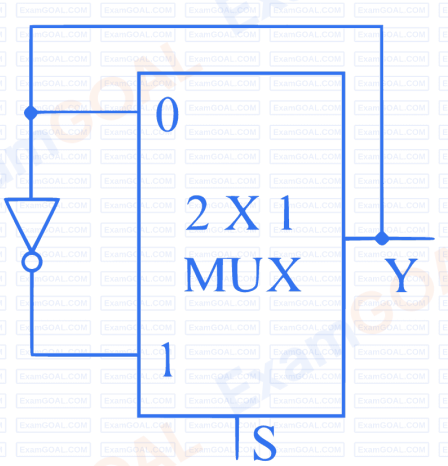
(C) $D_3\overline{D_2} + \overline{D_3}D_1$ (D) $D_3 + \overline{D_2}D_1$

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Q19. GATE ECE 2024

Open solution

The propagation delay of the 2 x 1 MUX shown in the circuit is 10 ns. Consider the propagation delay of the inverter as 0 ns.



If S is set to 1 then the output Y is _____.

(A) a square wave of frequency 100 MHz

(B) a square wave of frequency 50 MHz

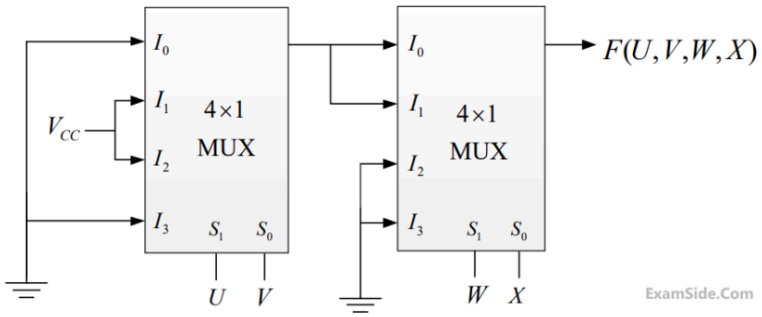
(C) constant at 0

(D) constant at 1

Q20. GATE ECE 2018

Open solution

A four-variable Boolean function is realized using 4 × 1 multiplexers as shown in the figure.



The minimized expression for F(U, V, W, X) is

(A) $\left(UV+\overline{UV}\right)\overline{W}$

(B) $\left(UV+\overline{UV}\right)\left(\overline{WX}+\overline{W}X\right)$

(C) $\left(U\overline{V}+\overline{U}V\right)\overline{W}$

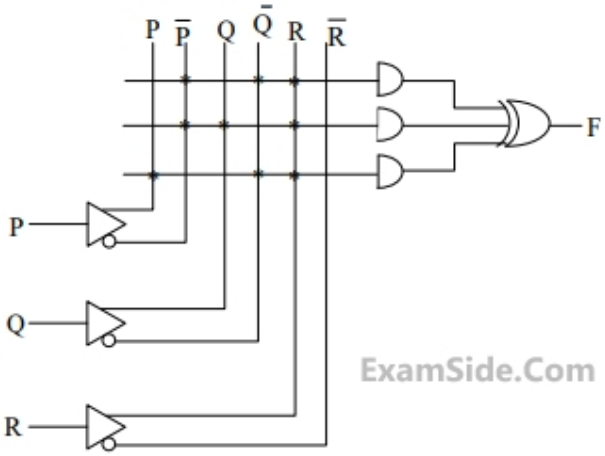
(D) $\left(U\overline{V}+\overline{U}V\right)\left(\overline{WX}+\overline{W}X\right)$

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Q21. GATE ECE 2017 Set 2

Open solution

A programmable logic array (PLA) is shown in the figure.



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The Boolean function F implemented is

(A) $\overline{P}\,\overline{Q}\,R + \overline{P}QR + P\overline{Q}\,\overline{R}$

(B) $(\overline{P} + \overline{Q} + R)(\overline{P} + Q + R) + (P + \overline{P} + \overline{R})$

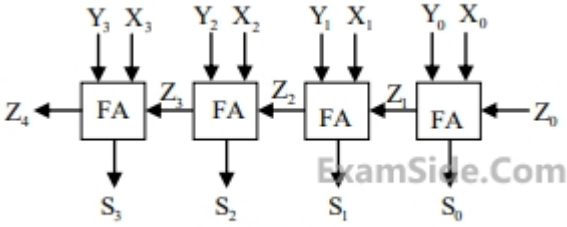
(C) $\overline{P}\,\overline{Q}\,R + \overline{P}QR + P\overline{Q}\,R$

(D) $(\overline{P} + \overline{Q} + R)(\overline{P} + Q + R) + (P + \overline{Q} + R)$

Q22. GATE ECE 2017 Set 2

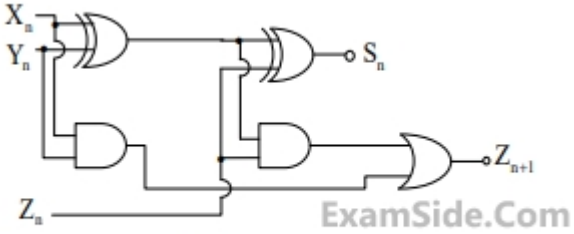
Open solution

Figure I shows a 4-bits ripple carry adder realized using full adders and Figure II shows the circuit of a full-adder (FA). The propagation delay of the XOR, AND and OR gates in Figure II are 20 ns, 15 ns and 10 ns respectively. Assume all the inputs to the 4-bit adder are initially reset to 0.



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Figure – I



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Figure – II

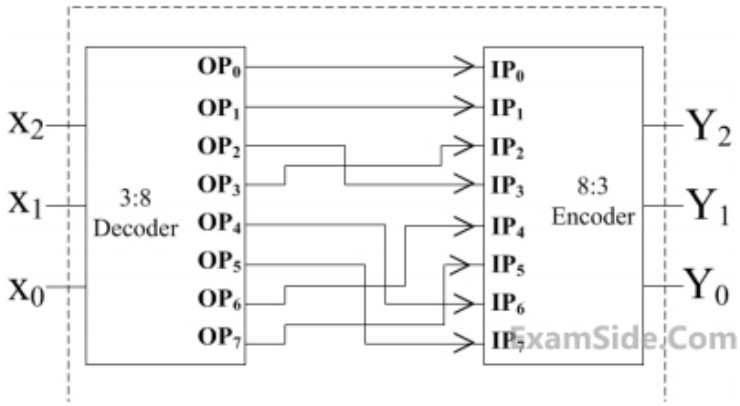
At t=0, the inputs to the 4-bit adder are changed to $X_3X_2X_1X_0$ =1100, $Y_3Y_2Y_1Y_0$ = 0100 and Z_0=1. The output of the ripple carry adder will be stable at t (in ns) = ____

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Q23. GATE ECE 2016 Set 1

Open solution

Identify the circuit below.



(A) Binary to Gray code converter

(B) Binary to XS3 converter

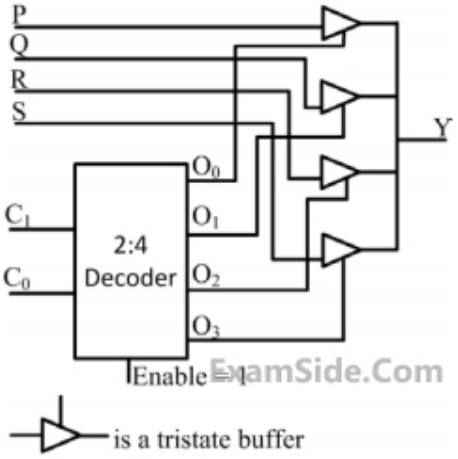
(C) Gray to Binary converter

(D) XS3 to Binary converter

Q24. GATE ECE 2016 Set 1

Open solution

The functionality implemented by the circuit below is



(A) 2-to-1 multiplexer

(B) 4-to-1 multiplexer

(C) 7-to-1 multiplexer

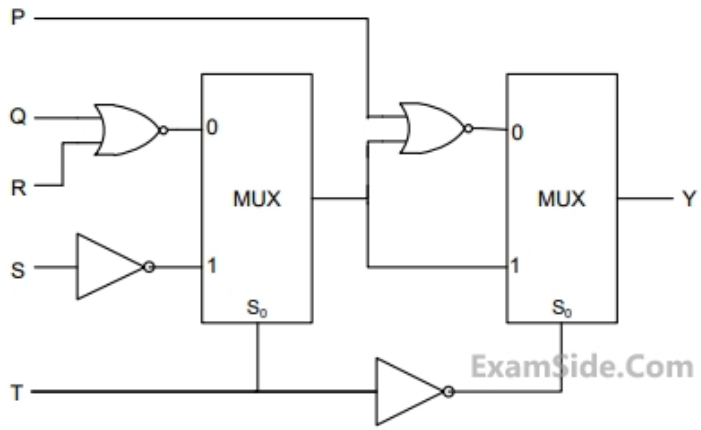
(D) 6-to-1 multiplexer

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Q25. GATE ECE 2016 Set 3

Open solution

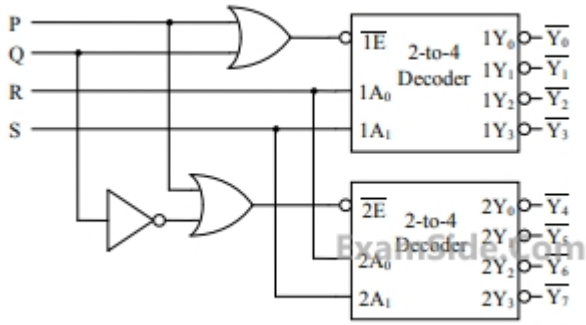
For the circuit shown in the figure, the delays of NOR gates, multiplexers and inverters are 2 ns, 1.5 ns and 1 ns, respectively. If all the inputs P, Q, R, S and T are applied at the same time instant, the maximum propagation delay (in ns) of the circuit is _____.



Q26. GATE ECE 2015 Set 2

Open solution

A 1-to-8 demultiplexer with data input D_{in} , address inputs S_0, S_1, S_2 (with S_0 as the LSB) and $\overline{Y}_0$ to $\overline{Y}_7$ as the eight demultiplexed outputs, is to be designed using two 2-to-4 decoders (with enable input $\overline{E}$ and address inputs A_0 and A_1) as shown in the figure. D_{in}, S_0, $S_1$and S_2 are to be connected to P, Q, R and S, but not necessarily in this order. The respective input connections to P, Q, R, and S terminals should be



(A) S_2, D_{in}, S_0, S_1

(B) S_1, D_{in}, S_0, S_2

(C) D_{in}, S_0, S_1 S_2

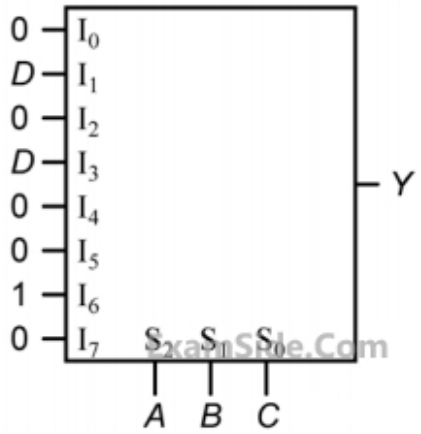
(D) D_{in}, S_2, S_0, S_1

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Q27. GATE ECE 2014 Set 4

Open solution

An 8-to-1 multiplexer is used to implement a logical function Y as shown in the figure. The output Y is given by



(A) Y= $A\overline{B}C + A\overline{C}D$

(B) $Y = \overline{A}BC + A\overline{B}D$

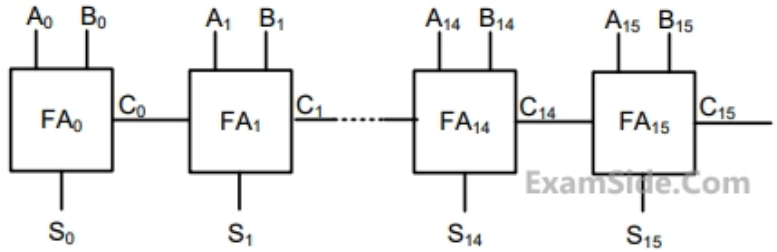
(C) $Y = AB\overline{C} + \overline{A}CD$

(D) $Y = \overline{A}\,\overline{B}D + A\overline{B}C$

Q28. GATE ECE 2014 Set 4

Open solution

A 16-bit ripple carry adder is realized using 16 identical full adders (FA) as shown in the figure. The carry-propagation delay of each FA is 12 ns and the sum-propagation delay of each FA is 15 ns. The worst case delay (in ns) of this 16-bit adder will be_____.

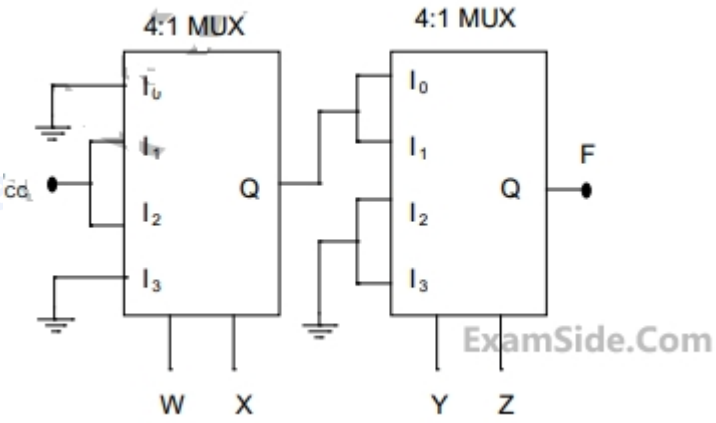


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Q29. GATE ECE 2014 Set 3

Open solution

In the circuit shown, WW and YY are MSBs of the control inputs. The output FF is given by



(A) $F = W\overline{X} + \overline{W}X + \overline{Y}\,\overline{Z}$

(B) $F = W\overline{X} + \overline{W}X + \overline{Y}\,Z$

(C) $F = W\overline{X}\,\overline{Y} + \overline{W}X\,\overline{Y}$

(D) $F = (\overline{W} + \overline{X})\,\overline{Y}\,\overline{Z}$

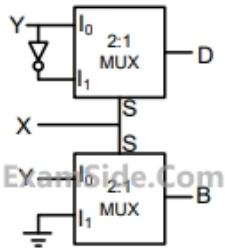
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Q30. GATE ECE 2014 Set 3

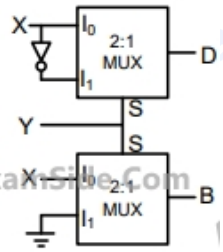
Open solution

If X and Y are inputs and the Difference (D = X − Y) and the Borrow (B) are the outputs, which one of the following diagrams implements a half-subtractor?

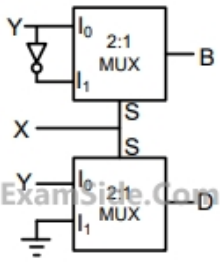
(A)



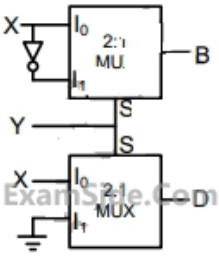
(B)



(C)



(D)

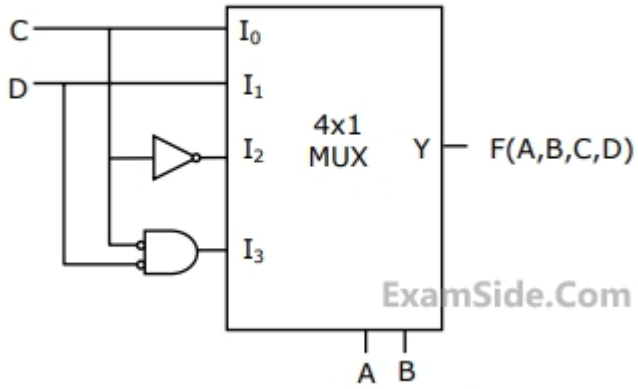


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Q31. GATE ECE 2010

Open solution

The Boolean function realized by the logic circuit shown is



(A) F=∑m(0,1,3,5,9,10,14)

(B) F=∑m(2,3,5,7,8,12,13)

(C) F=∑m(1,2,4,5,11,14,15)

(D) F=∑m(2,3,5,7,8,9,12)

Q32. GATE ECE 2009

Open solution

What are the minimum number of 2-to 1 multiplexers required to generate a 2-input AND gate and a 2-input EX-OR gate?

(A) 1 and 2

(B) 1 and 3

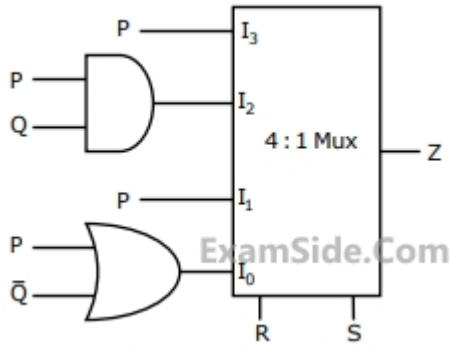
(C) 1 and 1

(D) 2 and 2

Q33. GATE ECE 2008

Open solution

For the circuit shown in the following figure I_0 - I_3 are inputs to the 4:1 multiplexer. R (MSB) and S are control bits. The output Z can be represented by



(A) PQ+P$\overline{Q}S + \overline{Q}\,\overline{R}\,\overline{S}$

(B) $P\overline{Q} + PQ\overline{R} + \overline{R}\,\overline{Q}\,\overline{S}$

(C) $P\overline{Q}\,\overline{R} + \overline{P}QR + PQRS + \overline{Q}\,\overline{R}\,\overline{S}$

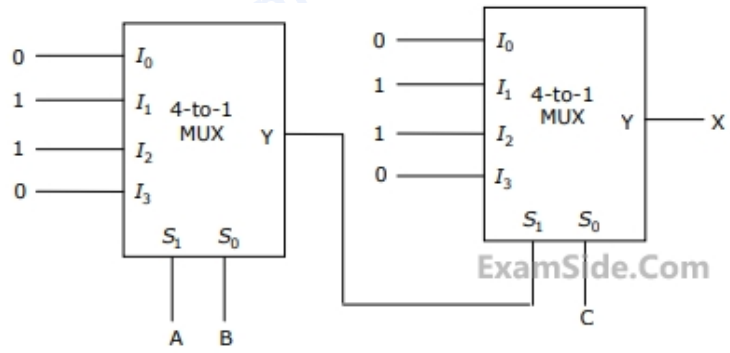
(D) $PQ\overline{R} + PQR\overline{S} + P\overline{Q}\,\overline{R}S + \overline{Q}\,\overline{R}\,\overline{S}$

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Q34. GATE ECE 2007

Open solution

In the following circuit, X is given by



(A) $X = A\overline{B}\,\overline{C} + \overline{A}\,B\,\overline{C} + \overline{A}\,\overline{B}C + ABC$

(B) $X = \overline{A}\,BC + A\overline{B}C + AB\overline{C} + \overline{A}\,\overline{B}\,\overline{C}$

(C) $X = AB + BC + AC$

(D) $X = \overline{A}\,\overline{B} + \overline{B}\,\overline{C} + \overline{A}\,\overline{C}$

Q35. GATE ECE 2004

Open solution

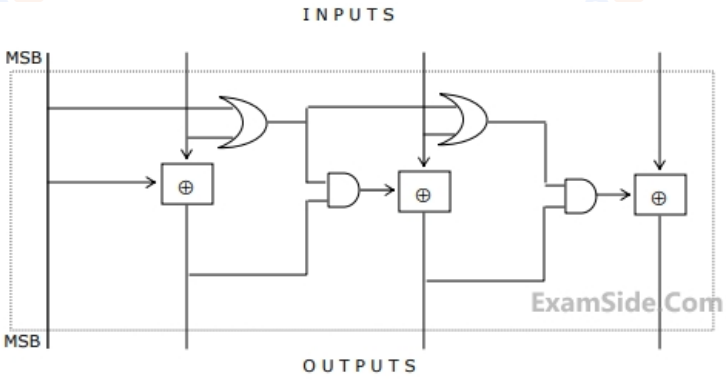
The minimum number of 2 to 1 multiplexers required to realize a 4 to 1 mutliplexer is

(A) 1 (B) 2 (C) 3 (D) 4

Q36. GATE ECE 2003

Open solution

The circuit shown in figure converts



(A) BCD to binary code

(B) Binary to excess – 3 code

(C) Excess – 3 to Gray code

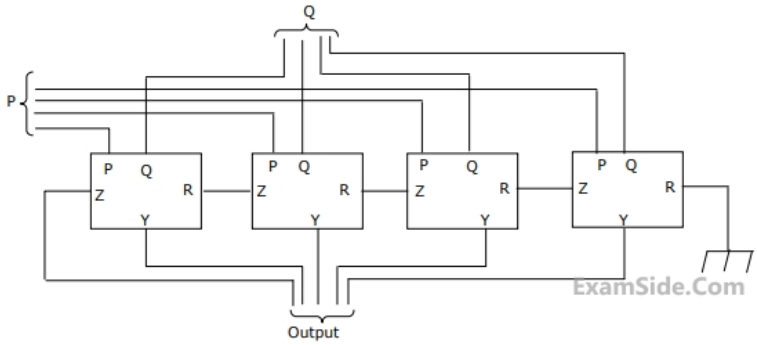
(D) Gray to Binary code.

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Q37. GATE ECE 2003

Open solution

The circuit shown in figure has 4 boxes each described by inputs P, Q, R and outputs Y, Z with
Y = $P \oplus Q \oplus R$
z= $RQ + \overline{P}R + Q\overline{P}$



The circuit acts as a

(A) 4 bit adder giving P + Q
(B) 4 bit subtractor-giving P - Q
(C) 4 bit subtractor-giving Q - P
(D) 4 bit adder giving P + Q + R

Q38. GATE ECE 2001

Open solution

In the TTL circuit in Figure 2.11, S_2 to S_0 are select lines and X_7 to X_0 are input lines. S_0 and X_0 are LSBs. The output Y is

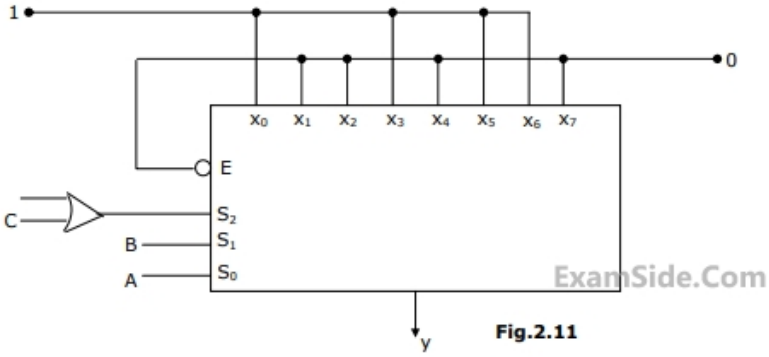


Fig.2.11

(A) indeterminate
(B) A ⊕ B
(C) $\overline{A \oplus B}$
(D) $\overline{C}\,\overline{(A \oplus B)} + C(A \oplus B)$

Q39. GATE ECE 1999

Open solution

For a binary half-sub-tractor having two inputs A and B, the correct set of logical expressions for the outputs D (=A minus B) and X (=borrow) are

(A) $D = AB + \overline{A}\,B,\ X = \overline{A}\,B$
(B) $D = \overline{A}\,B + A\overline{B} + A\overline{B},\ X = A\overline{B}$
(C) $D = \overline{A}\,B + A\overline{B},\ X = \overline{A}\,B$
(D) $D = AB + \overline{A}\,\overline{B},\ X = A\overline{B}$

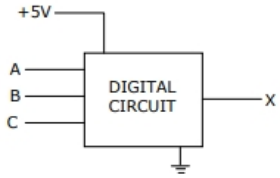
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Marks 5 1 questions

Q40. GATE ECE 2002

Open solution

The inputs to a digital circuit shown in Figure 9(a) are the external signals A, B and C.



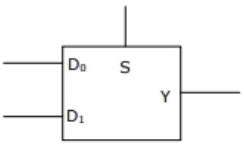


Table 9

S	Y
0	D_0
1	D_1

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($\overline{A}\,\overline{B}$ and $\overline{C}$ and are not available). The +5V power supply (logic 1) and the ground (logic 0) are also available. The output of the circuit is X = $\overline{A}\,B + \overline{A}\,\overline{B}\,\overline{C}$

(a) Write down the output function in its canonical SOP and POS forms.

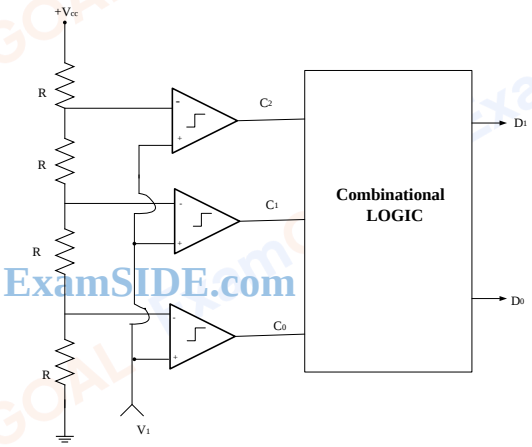
(b) Implement the circuit using only two 2:1 multiplexers shown in the Figure where S is the data-select line, D_0 and D_1 are the input data lines and Y is the output lines. The function table for the multiplexer is given in table

Marks 8 1 questions

Q41. GATE ECE 1987

Open solution

The circuit diagram of a 2 bit A to D converter is shown in figure below. The combinational logic is to be disigned to provide a natural binary representation using D_1 and D_0 for the analog input V_i. D_1 is to be the most significant bit.



(a) Draw the Karnaugh maps for D_1 and D_0 in terms of C_2, C_1 and C_0

(b) Obtain the minimal sum of products expressions for D_1 and D_0.

(c) Realize the logic for D_1 and D_0 using 2- input NAND gates only.

(d) Find the resolution of the Ato D converter.

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Marks 10 4 questions

Q42. GATE ECE 1995

A ROM is to be used to implement the Boolean functions given below:
$F_1(A,\ B,\ C,\ D) = ABCD + \bar{A}\,\overline{B}\,\bar{C}\,\bar{D}$
$F_2(A,\ B,\ C,\ D) = (A + B)(\bar{A} + \overline{B} + C)$
$F_3(A,\ B,\ C,\ D) = \sum 13, 15 + \sum 3, 5$
(a) What is the minimum size of the ROM required?

(b) Determine the data in each location of the ROM.

Open solution

Q43. GATE ECE 1994

A Boolean function, F , given as sum of product (SOP) terms as F= $\sum$m(3,4,5,6) with A,B, and C as inputs. The function, F, can be expressed on the Karnaugh's map shown below.
(1) What will be the minimized SOP expression for F?
(2) Implement this function on an 8 : 1 MUX.

Open solution

Q44. GATE ECE 1993

Signals A,B,C,D and $\overline{D}$ are available. Using a single 8 - to - 1 multiplexer and no other gate, implement the Boolean function.

$f(A, B, C, D) = B.C + A.B.\bar{D} + \bar{A}.\bar{C}.\bar{D}$

Open solution

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Q45. GATE ECE 1989

Open solution

A chemical reactor has three sensors indicating the following conditions:-
(1) Pressure (P) is low or high'
(2) Temperature (T) is low or high' and
(3) Liquid level (L) is low or high.
its has two controls - Heater (H) which is either on or off and inlet value (V) which is open or close. The controls are operated as per Table.

(a) Using the convertion High =1, Low = 0, On=1, Off=0, Open=1 and Closed=0, draw the Karnaugh maps for H and V.

(b) Obtain the minimal product of sums expressions for H and V.

(c) Realize the logic for H and V using two 4-input multiplexers with T and L as control inputs. Used T as MSB.

Inputs			Output	
P	T	L	H	V
LOW	LOW	LOW	Off	Open
LOW	LOW	High	On	Closed
LOW	High	LOW	Off	Open
LOW	High	High	Off	Closed
High	LOW	LOW	On	Open
High	LOW	High	On	Closed
High	High	LOW	Off	Closed
High	High	High	x	x

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Answer Key

Marks 1

1. **B**	2. **A**	3. **C**	4. **B**	5. **D**
6. **B**	7. **A**	8. **C**	9. **B**	10. **D**
11. **A**	12. **C**	13. **B**	14. **D**	15. **B**
16. **B**				

Marks 2

17. **C**	18. **B**	19. **B**	20. **C**	21. **A**
22. **50**	23. **A**	24. **B**	25. **6**	26. **D**
27. **C**	28. **195**	29. **C**	30. **A**	31. **D**
32. **A**	33. **A**	34. **A**	35. **C**	36. **D**
37. **B**	38. **B**	39. **C**		

Marks 5

40. **a) X=**
$\overline{A}BC + \overline{A}B\overline{C} + \overline{A}\,\overline{B}\,\overline{C}$

$\sum m(0,2,3) - SOP$

$X = \pi(1,4,5,6,7) - POS$

Marks 8

41. **(b)** D_1**=**C_2
D_0**=** $\overline{C_2}C_1$
(d) Resolution =
$\frac{V_{CC}}{3}$

Marks 10

42. **a)Size of ROM=48**	43. **1)** $F = \overline{A}B + A\overline{C} + \overline{A}BC$	44. **Solve it.**	45. **(b)** $H = \overline{T}(P + L)$ $V = \overline{L}(\overline{T} + \overline{P})$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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